ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

C01-AT1

EXPERIMENT 1

AIM:

To write a MATLAB program for transformation between Cartesian, Cylindrical and Spherical coordinate systems.

PROGRAM:

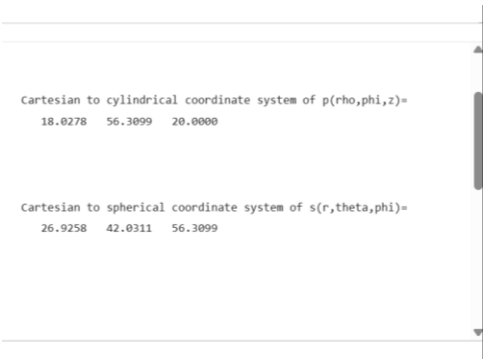
```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
% Cartesian to Cylindrical
rho=sqrt(x^2+y^2);
phi=atan2d(y,x);
disp('Cartesian to cylindrical coordinate system of p(rho,phi,z)=');
disp([rho phi z]);
% Cartesian to Spherical
r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
phi=atan2d(y,x);
disp('Cartesian to spherical coordinate system of s(r,theta,phi)=');
disp([r theta phi]);
```

```
clc;
rho=input('Enter the value of rho=');
phi=input('Enter the value of phi=');
z=input('Enter the value of z=');
```

```
% Cylindrical to Cartesian
x=rho*cosd(phi);
y=rho*sind(phi);
disp('Cylindrical to Cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Cylindrical to Spherical
r=sqrt(rho^2+z^2);
theta=acosd(z/r);
disp('Cylindrical to spherical coordinate system of s(r,theta,phi)=');
disp([r theta phi]);

clc;
r=input('Enter the value of r=');
theta=input('Enter the value of theta (deg)=');
phi=input('Enter the value of phi (deg)=');
% Spherical to Cartesian
x=r*sind(theta)*cosd(phi);
y=r*sind(theta)*sind(phi);
z=r*cosd(theta);
disp('Spherical to Cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Spherical to Cylindrical
rho=r*sind(theta);
disp('Spherical to cylindrical coordinate system of s(rho,phi,z)=');
disp([rho phi z]);
```

OUTPUT:




```
Cylindrical to Cartesian coordinate system of p(x,y,z)=
    9.6593    2.5882   20.0000


Cylindrical to spherical coordinate system of s(r,theta,phi)=
   22.3607   26.5651   15.0000
```

```
Spherical to Cartesian coordinate system of p(x,y,z)=
    2.4321    0.8852    9.6593

Spherical to cylindrical coordinate system of s(rho,phi,z)=
    2.5882   20.0000    9.6593
```

RESULTS:

Thus the transformation between Cartesian, Cylindrical and Spherical coordinate systems have been done using MATLAB software.

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Apply coordinate-system definitions and angle conventions	Apply coordinate-system definitions, symbols, units, and angle conventions	Applies coordinate-system definitions and angle conventions throughout.	Uses correct definitions with one minor notation or convention error.	Shows partial understanding; multiple corrections are needed.	Uses incorrect coordinate definitions or angle conventions.	25	2.5	
2	Implement transformations in MATLAB	Implement six transformations in MATLAB	Implements all six transformations correctly using suitable degree-based functions.	Implements most transformations correctly with minor coding errors.	Implements some transformations but needs assistance to correct errors.	Unable to implement the required transformations.	25	2.5	
3	Verify numerical values and reverse transformations	Verify numerical values and reverse transformations	All outputs agree with analytical values; reverse transformations recover the original point within rounding precision.	Most values agree; small unexplained differences occur.	Several values differ or reverse verification is incomplete.	Outputs are incorrect or not verified.	30	3	
4	Interpret and document results	Interpret and document results	Presents complete tables, units, deviations,	Documentation is clear with minor omissions.	Tables or interpretation are incomplete.	Results and conclusion are missing or unclear.	20	2	

			and a clear conclusion.						
Total							100	10	